

VISION Resistivity

Dual Frequency

Recorded Mode, Run 10



Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

UWID:

Longitude: 3° 26' 36.13" E

Rig Name:

West Hercules

Block: 31/5

Rig Type:

Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 8 1/2 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level



Ground Level: 307.00 m

Acquisition Dates: 31-Dec-2019 -- 01-Jan-2020

Other Services:

Log Interval: 2574.00(m) -- 2644.00(m)

Direction & Inclination (Surveys)

Index Types: Measured Depth

Annular Pressure While Drilling

Index Scales: 1:200

geoVISION Bit Resistivity

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019

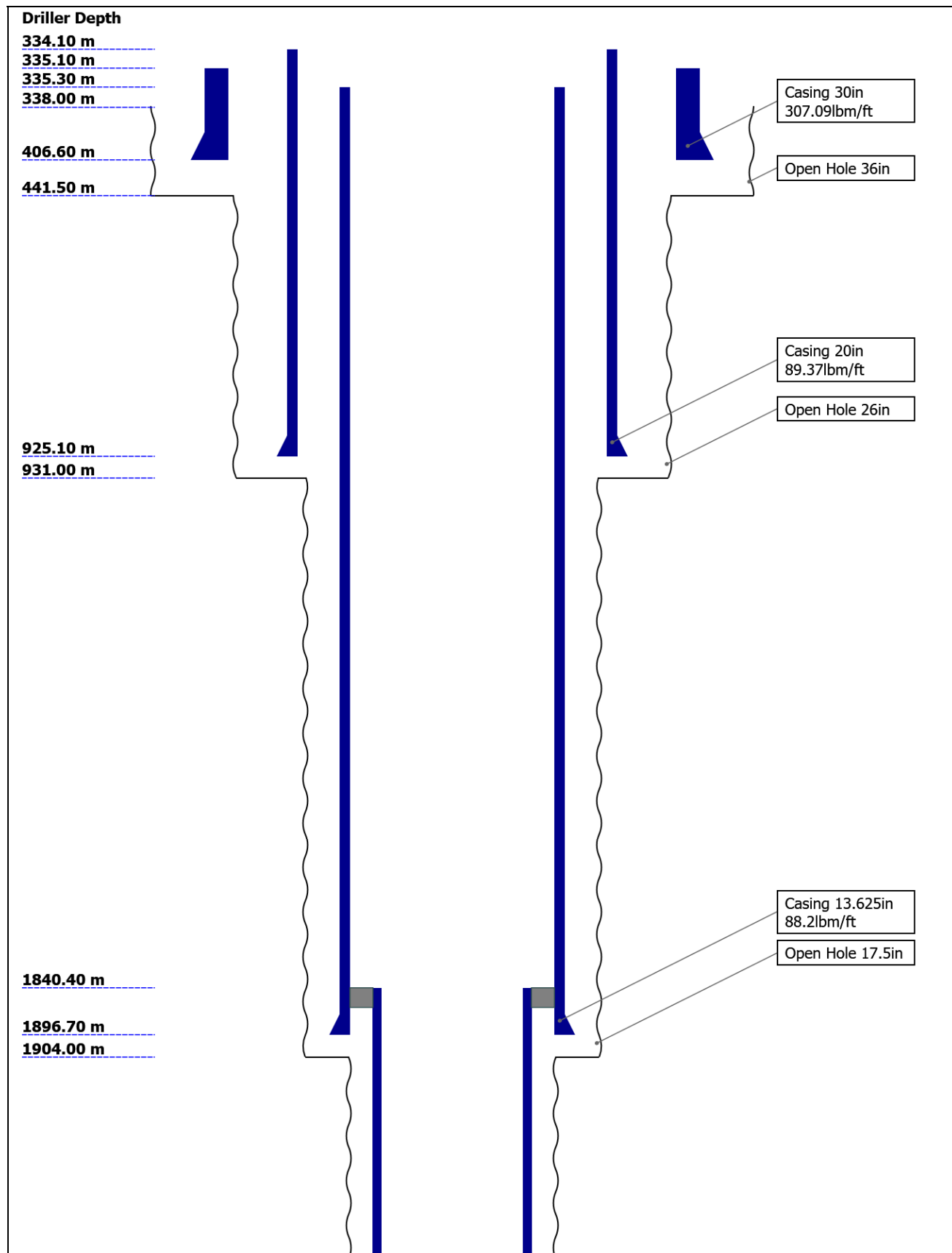
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Well Sketch



2583.00 m

2584.00 m

2643.00 m

Casing 9.625in

53.5lbm/ft

Open Hole 12.25in

Open Hole 8.5in

Borehole Size/Casing Record

Bit						
Bit Size (in)	36	26	17.5	12.25	8.5	
Top Driller (m)	338	441.5	931	1904	2584	
Bottom Driller (m)	441.5	931	1904	2584	2643	
Casing						
Size (in)	30	20	13.625	9.625		
Weight (lbm/ft)	307.09	89.37	88.2	53.5		
Inner Diameter (in)	27.067	18.73	12.375	8.535		
Grade	X56	N80	P110	P110		
Top Driller (m)	335.1	334.1	335.3	1840.4		
Bottom Driller (m)	406.6	925.1	1896.7	2583		

Operational Run Summary

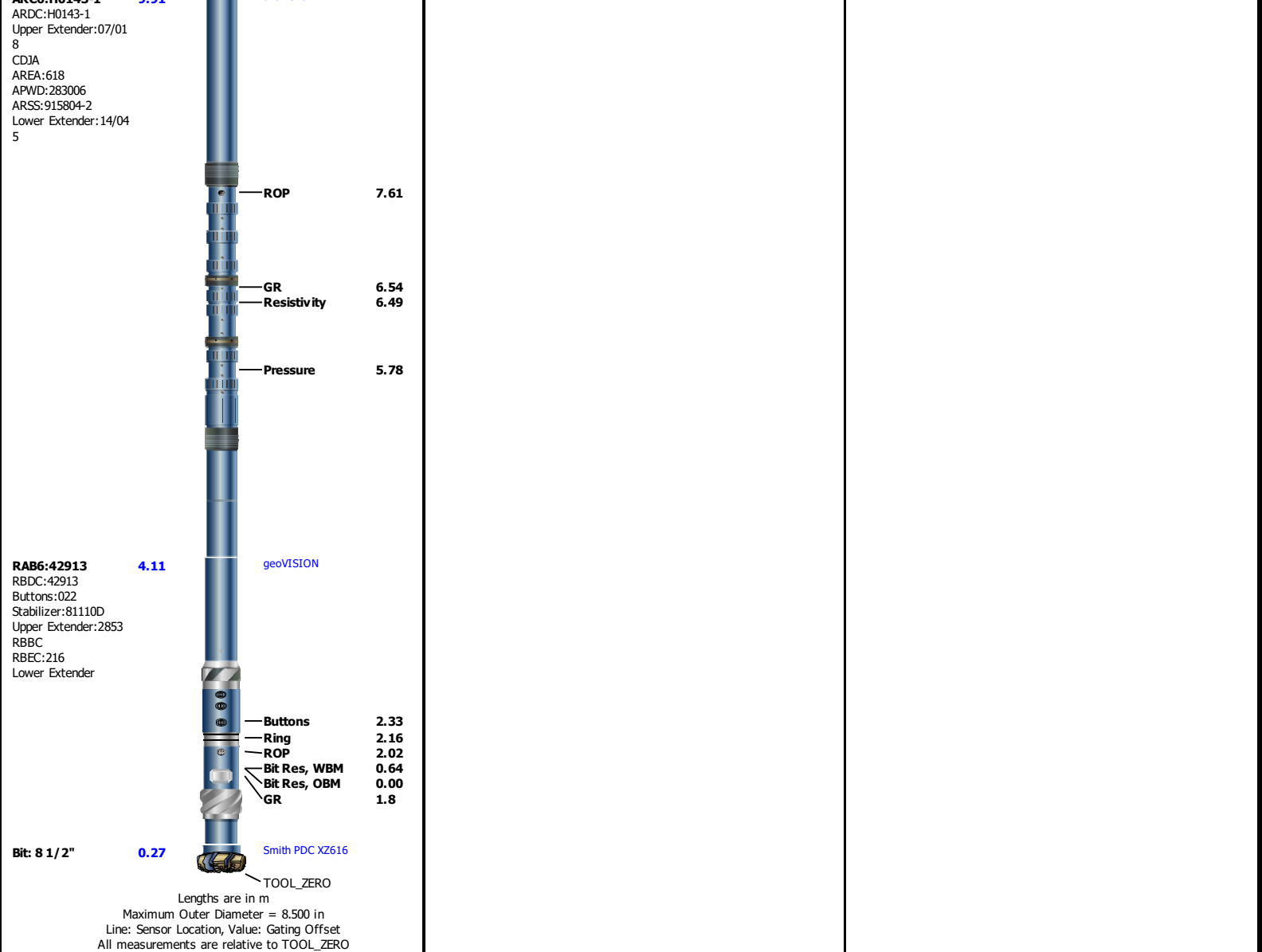
Parameter (unit)	Run 10					
Date Log Started	31-Dec-2019					
Time Log Started	17:32:56					
Date Log Finished	01-Jan-2020					
Time Log Finished	07:38:34					
Bit Size (in)	8.500					
Bit Start Depth (m)	2596.00					
Bit Stop Depth (m)	2643.00					
Top Log Interval (m)	2584.90					
Bottom Log Interval (m)	2642.28					
Max Hole Deviation (deg)	1.12					
Azimuth of Max Deviation (deg)	120.28					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	S.Kroevel					
Witnessed By	J.Eikenes					
Service Order Number	19SCA0085					

Borehole Fluids

Parameter(unit)	Run 10					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	43					
Source of Sample	Active Tank					
Salinity (ppm)	93984					
Density (g/cm3)	1.15					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.07 @ 15					
RMF @ Meas Temp (ohm.m@degC)	0.06 @ 20					
RMC @ Meas Temp (ohm.m@degC)	0.09 @ 15					
RM @ BHT (ohm.m@degC)	0.04 @ 43					
RMF @ BHT (ohm.m@degC)	0.04 @ 43					
RMC @ BHT (ohm.m@degC)	0.05 @ 43					
Total Solid (%)						
High Gravity Solids (%)						

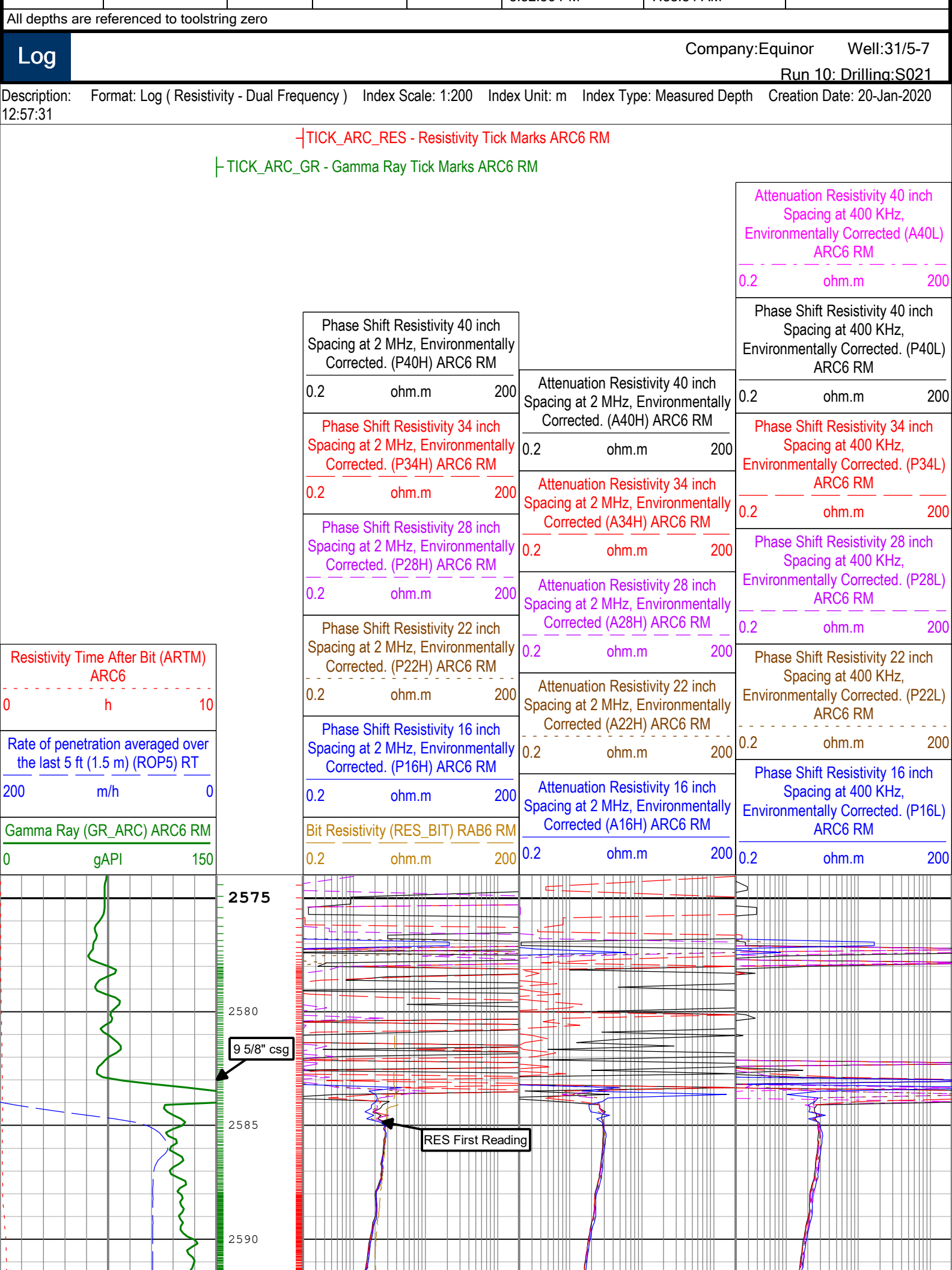
Remarks and Equipment Summary

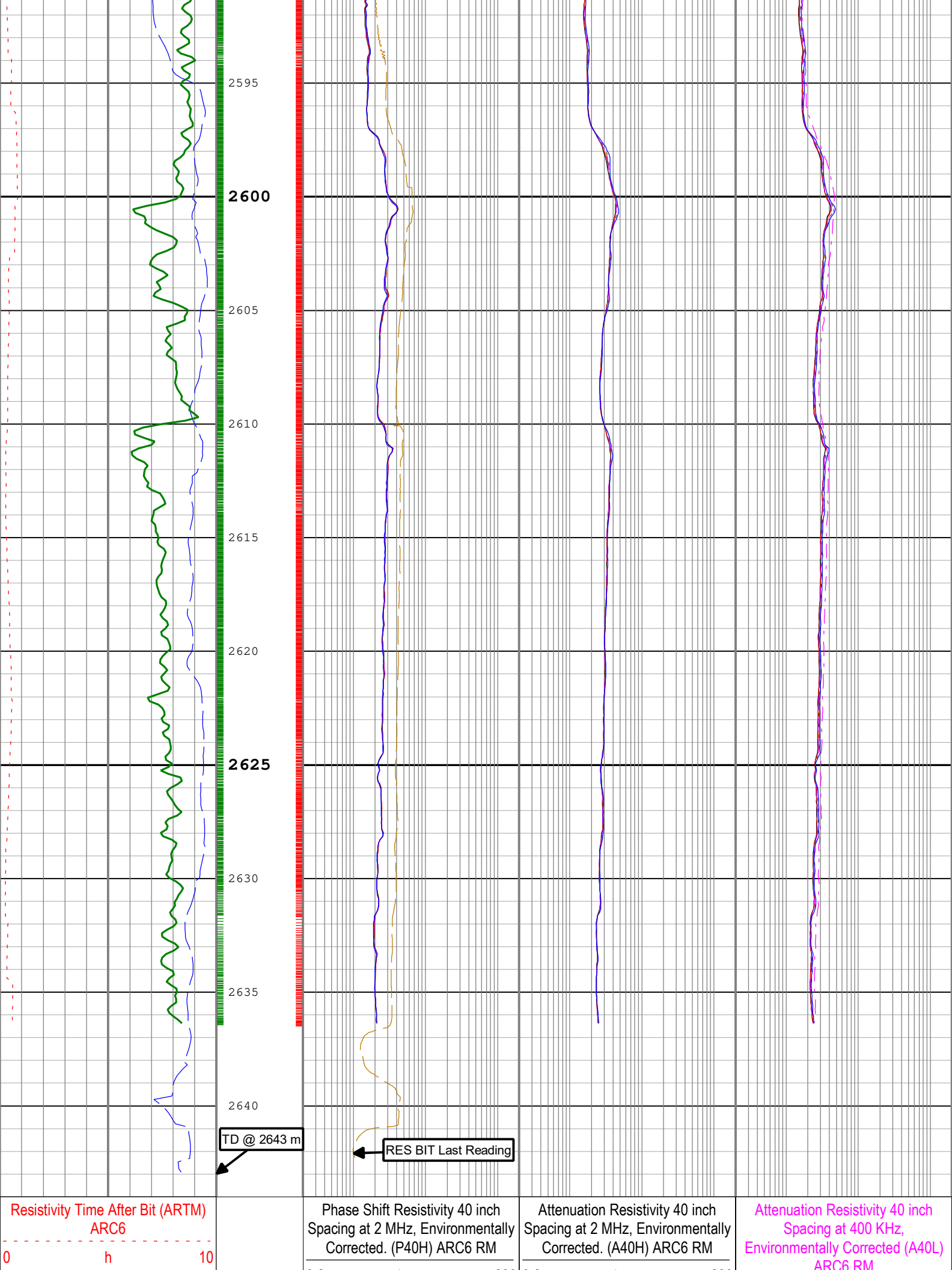
Run 10: Toolstring				Run 10: Remarks	
Equip name TELE675:E4388 Length 18.37 MP name TeleScope Offset MSSU675:30848 Upper Extender MDC675:E4388 MMA:2507 MDI:3666 PMVC:236 PMEA:803 MTA:2162 MTK675:004 MSSD675:1005482-1 Lower Extender, Long  D&I 14.14 Vibration 13.14 ROP 11.79 ARC6:H0143-1 9.91 arcVISION	All depths referenced to driller's depth. See EOWR for depth tracking details.				
	All data from tool memory. All data acquired while drilling.				
	Gamma Ray measurement is environmentally corrected for mud weight, bit size, collar thickness and Potassium content.				
	Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.				
	Data acquired while reaming down over cored interval from 2592 to 2596 m bit depth.				
	Run 10 POOH at 2643 m to perform coring.				



Run 10			
VISION Resistivity - Dual Frequency			
Software Version			
Acquisition System		Version	
Maxwell 2019.2		9.2.113335.3100	
Application Patch		DnM_TestKit-XceedCPU41-2019.2_9.2.114181	
Computation	Description		Version
ARCResistivity	ARC Resistivity Computation Package for ARC Tool Family		9.2.113335.3100
ARC6GammaRay	ARC6 Gamma Ray Computation Package for both Real-time and Recorded Mode		9.2.113335.3100
GVRRes	GVR Resistivity Computation Package for both Real-time and Recorded Mode		9.2.113335.3100
Tool Interface	System Version	Loaded Version	
HSPM	2019.0c.001	9.2.113335.3100	
Tool Elements	Description	Software Version	Firmware Version
RBEC	Electronics Chassis Assembly for RAB6-C	9.2.113335.3100	V8.8B
DRILLING_SURFACE	DRILLING_SURFACE	9.2.113335.3100	
ARDC	ARC 6.75 Inch Tool Drill Collar	9.2.113335.3100	V9.7B

Pass Summary							
Run Name	Pass Objective	Direction	Top	Bottom	Start	Stop	Include Parallel Data
Run 10	Drilling	Down	236.78 m	2642.97 m	31-Dec-2019 5:32:56 PM	01-Jan-2020 7:38:34 AM	Yes





Resistivity Time After Bit (ARTM)
ARC6

0 h 10

Phase Shift Resistivity 40 inch
Spacing at 2 MHz, Environmentally
Corrected. (P40H) ARC6 RM

Attenuation Resistivity 40 inch
Spacing at 2 MHz, Environmentally
Corrected. (A40H) ARC6 RM

Attenuation Resistivity 40 inch
Spacing at 400 KHz,
Environmentally Corrected (A40L)
ARC6 RM

Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT 200m/h0 Gamma Ray (GR_ARC) ARC6 RM 0gAPI150	0.2ohm.m200		0.2ohm.m200		0.2ohm.m200	
	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC6 RM		Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC6 RM		Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC6 RM	
	0.2ohm.m200		0.2ohm.m200		0.2ohm.m200	
	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC6 RM		Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC6 RM		Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC6 RM	
	0.2ohm.m200		0.2ohm.m200		0.2ohm.m200	
	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC6 RM		Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC6 RM		Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC6 RM	
	0.2ohm.m200		0.2ohm.m200		0.2ohm.m200	
	Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC6 RM		Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC6 RM		Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC6 RM	
	0.2ohm.m200		0.2ohm.m200		0.2ohm.m200	
	Bit Resistivity (RES_BIT) RAB6 RM				Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC6 RM	
0.2ohm.m200				0.2ohm.m200		
┆TICK_ARC_GR - Gamma Ray Tick Marks ARC6 RM ┆TICK_ARC_RES - Resistivity Tick Marks ARC6 RM						
Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:200 Index Unit: m Index Type: Measured Depth Creation Date: 20-Jan-2020 12:57:31						

Channel Processing Parameters				
Run 10: Parameters				
Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC6	NO_TX_FAILED	
BH_COMPUTE	Borehole Effect Computation Option	ARC6	No	
BHK	Drilling Fluid Potassium Concentration	Borehole	6.3	%
BHT	Bottom Hole Temperature	Borehole	43	degC
BS	Bit Size	DNMSESSION	Depth Zoned	in
DEPTH_SEL	Depth Selection Parameter	DNMSESSION	Driller's Depth	
DFD	Drilling Fluid Density	Borehole	1.15	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
ERRCT	Percentage Error Cutoff	ARC6	4.5	%
GRSE_RM	Generalized Mud Resistivity Selection for Recorded Mode	Borehole	REMS(RM)	
GRSH	Gamma Ray Shale	ARC6	1000	gAPI
GTSE_RM	Generalized Temperature Selection for Recorded Mode	Borehole	DHAT(RM)	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC6	2	ohm.m
INVAS_COMPUTE	Invasion Computation Option	ARC6	No	
LOW_BLEND	Low Resistivity Threshold for Blending	ARC6	1	ohm.m
MST	Mud Sample Temperature	Borehole	15	degC
MSWS	ARCWizard Model Switch Window Size	ARC6	152.4	cm
MULTIEFFECT_COMPUTE	Multi-effect Computation Option	ARC6	No	

PRTD	ARCWizard Preferred Resistivity Log for Rt Display while Multi-Effects	ARC6	P34B	
RMS	Resistivity of Mud Sample	Borehole	0.07	ohm.m
T1WM	ARCWizard Weight Multiplier to Measurements for Transmitter 1	ARC6	1	
T2WM	ARCWizard Weight Multiplier to Measurements for Transmitter 2	ARC6	1	
T3WM	ARCWizard Weight Multiplier to Measurements for Transmitter 3	ARC6	1	
T4WM	ARCWizard Weight Multiplier to Measurements for Transmitter 4	ARC6	1	
T5WM	ARCWizard Weight Multiplier to Measurements for Transmitter 5	ARC6	1	
ATMP_ARC	ARC Temperature Selection	ARC6	Annular	
TEMP_SEL_RAB	RAB Temperature Selection	RAB6	Tool	
UNIFORM_COMPUTE	Uniform Rock Computation Option	ARC6	No	

Depth Zone Parameters			
Parameter	Value	Start (m)	Stop (m)
BS	12.25	2574	2584
BS	8.5	2584	2643
All depth are actual.			

Tool Control Parameters	
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Run 10: Parameters				
Parameter	Description	Tool	Value	Unit
OFFBTM_TH	Threshold for deciding whether the bit is off bottom	DNMSESSION	0	m

Calibration Report	
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RAB6 (GeoVision Resistivity 675) Calibration - Run 10			
Primary Equipment :			
Electronics Chassis		RBEC	216

M21V - Resistivity							
Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 2 at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

M22V - Resistivity							
Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 2 at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250	

M01V - Resistivity							
Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 0 at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9950	1.0250	

M02V - Resistivity							
Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Monitor 0 at T2 Calibration Coefficient		Master	1.0000	0.9750	0.9980	1.0250	

R1V - Resistivity							
Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Ring at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

R2V - Resistivity							
Master (Time Frame File):		16:32:26 08-Aug-2019					
Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Ring at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250	

BDM1 - Resistivity							
Master (Time Frame File):		16:32:26 08-Aug-2019					

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Deep at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0070	1.0250	

BDM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Deep at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0100	1.0250	

BMM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Medium at T1 Calibration Coefficient		Master	1.0000	0.9750	0.9990	1.0250	

BMM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Medium at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0020	1.0250	

BSM1 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Shallow at T1 Calibration Coefficient		Master	1.0000	0.9750	1.0000	1.0250	

BSM2 - Resistivity

Master (Time Frame File): 16:32:26 08-Aug-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Button Shallow at T2 Calibration Coefficient		Master	1.0000	0.9750	1.0030	1.0250	

PGR - Gamma Ray: Blanket

Master (Time Frame File): 14:06:21 18-Dec-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Gamma Ray Calibration Gain		Master	1.0000	0.7500	0.9838	1.2500	

ARC6 (Array Resistivity Compensated 675) Calibration - Run 10

Primary Equipment :

Elec. Chassis HP w/o AIM Receiver

AREA

618

RESAIRCAL - Resistivity: Air

Master (Time Frame File): 15:26:50 07-Dec-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	7.951	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.997	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	4.584	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.904	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.142	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	1.927	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-1.871	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	1.878	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-1.879	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	1.856	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	7.970	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	6.989	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	4.593	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	4.889	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.160	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	-0.666	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	0.584	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	-0.629	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	0.582	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	-0.643	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:23:30 06-Jun-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit		
Gamma Ray Calibration Gain		Master	1.000	0.580	1.100	1.250		

Company: Equinor
Well: 31/5-7
Field: Eos
Rig Name: West Hercules
Country: Norway



Schlumberger

VISION Resistivity
Dual Frequency
Recorded Mode, Run 10